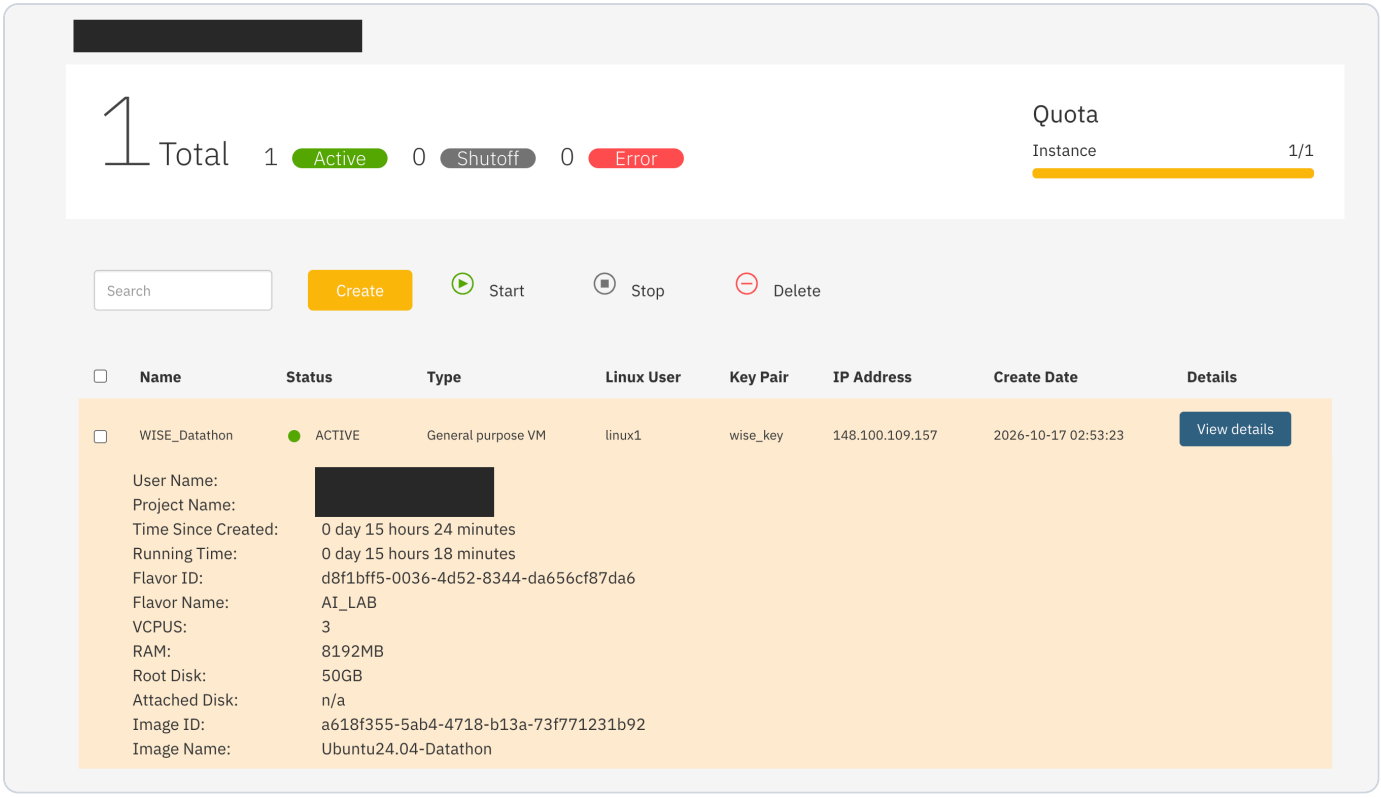
1 LinuxONE Community Cloud: VM instance

Instance WISE_Datathon is ACTIVE: 3 vCPUs, 8 GB RAM, 50 GB disk, Ubuntu 24.04.



2 Instance check

Architecture s390x, Big Endian, 3 vCPUs, 8 GB RAM.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="1" scripts/demo_ibm.sh --auto


== Step 1: The machine: IBM LinuxONE, s390x, big-endian
s390x
Architecture:                         s390x
Byte Order:                           Big Endian
vCPUs: 3
               total        used        free      shared  buff/cache   available
Mem:            8009        5660        1280          23        1229        2348
```

3 Container and services

IBM-provided JupyterLab container running. SecondLook API and Granite services active.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="2" scripts/demo_ibm.sh --auto


== Step 1: The platform: containers + services (Granite runs in the IBM-provided contai
ner)
container: notebook  image: registry.linuxone.cloud.marist.edu/l1cc/jupyterlab-image-s3
90x:latest
secondlook-api: active
secondlook-granite: active
```

4 API service status

systemd service active (running) with 3 uvicorn workers.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % ssh -i wise_key.pem linux1@148.100.109.157 'systemctl status secondlook-api --no-pager
  | head -12'

● secondlook-api.service - SecondLook API
     Loaded: loaded (/etc/systemd/system/secondlook-api.service; enabled; preset: enabled)
     Active: active (running) since Sat 2026-10-17 15:37:33 UTC; 2h 13min ago
   Main PID: 19260 (uvicorn)
      Tasks: 44 (limit: 9567)
     Memory: 312.8M (peak: 313.6M)
        CPU: 1min 47.229s
     CGroup: /system.slice/secondlook-api.service
             ├─19260 /home/linux1/secondlook/.venv/bin/python /home/linux1/secondlook/.venv/bin/uvicorn src.api.main:app --host 0.0.0.0
 --port 8000 --workers 3
             ├─19264 /home/linux1/secondlook/.venv/bin/python -c "from multiprocessing.resource_tracker import main;main(6)"
             ├─19265 /home/linux1/secondlook/.venv/bin/python -c "from multiprocessing.spawn import spawn_main; spawn_main(tracker_fd=7,
 pipe_handle=9)" --multiprocessing-fork
             ├─19266 /home/linux1/secondlook/.venv/bin/python -c "from multiprocessing.spawn import spawn_main; spawn_main(tracker_fd=7,
 pipe_handle=13)" --multiprocessing-fork
```

5 Resource usage

VM memory, and CPU and memory of the IBM container.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % ssh -i wise_key.pem linux1@148.100.109.157 'free -m; docker stats --no-stream'

               total        used        free      shared  buff/cache   available
Mem:            8009        5661        1278          23        1229        2347
Swap:              0           0           0
CONTAINER ID   NAME       CPU %     MEM USAGE / LIMIT     MEM %     NET I/O   BLOCK I/O        PIDS
61eee76a88dc   notebook   0.13%     3.977GiB / 7.821GiB   50.84%    0B / 0B   3.17GB / 277MB   97
```

6 API health

Service reports arch s390x. 37,491 applications, 2,600 denied, 1,726 queued.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="3" scripts/demo_ibm.sh --auto


== Step 1: The API on LinuxONE
{"status":"ok","arch":"s390x","python":"3.12.3","model_load_s":0.151,"version":"secondlook-0.1"}
applications: 37491 | denied: 2600 | queued: 1726
```

7 Live request (simulated application)

Matched against 1,915 approved candidates in about 3 ms on the server.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="4" scripts/demo_ibm.sh --auto


== Step 1: A live request (SIMULATED incoming application) - matching runs on this machine
SIMULATION: True | candidates: 1915 | server compute ms: 3.107
  twin 1 strong dist 0.0 income 141.0 loan 615000.0
  twin 2 strong dist 0.18 income 146.0 loan 595000.0
  twin 3 strong dist 0.23 income 135.0 loan 595000.0
round trip on the VM: 0.006178s
```

8 Performance under load

Median 3.9 ms for one client. 470 and 466 requests per second at 4 and 16 clients.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="5" scripts/demo_ibm.sh --auto


  == Step 1: Measured performance (300 requests, then concurrent clients)
  one client:  median 3.9 ms | p95 7.5 ms | 223 requests/s
   4 clients:  median   7 ms | p95  16 ms | 470 requests/s
  16 clients:  median  29 ms | p95  70 ms | 466 requests/s
```

9 IBM Granite 4.0 and zDNN

Granite served by llama.cpp. zDNN backend initialises; offload gave no speed-up, so no acceleration is claimed.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="6" scripts/demo_ibm.sh --auto


== Step 1: IBM Granite 4.0 (generative AI) served by llama.cpp inside the container
{"model": "granite-4.0-h-350m-base-be.f16.gguf", "load_s": 0.57, "n_ctx": 2048}
load_tensors:          zDNN model buffer size =   501.47 MiB
ggml_zdnn_init: found 1 device
ggml_zdnn_init: NNPA name: zDNN
```

10 Security gate

Demographic audit endpoint returns HTTP 403 without a token. API reached through an SSH tunnel.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % STEPS="7" scripts/demo_ibm.sh --auto


== Step 1: Security: demographic audit endpoint is token-gated
fairness-report WITHOUT token -> HTTP 403
SecondLook API/Granite are not exposed to the internet; reached through an SSH tunnel
```

11 Matching on the VM (s390x)

37,491 applications, three matching scopes, result digests shown.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % ssh -i wise_key.pem linux1@148.100.109.157 'cd ~/secondlook && .venv/bin/python -I -c
"import sys;sys.path.insert(0,\".\");exec(open(\"scripts/remote_matching.py\").read())" 2>&1 | grep -v Warn'

arch s390x rows 37491
lender: 0.65s pairs=7621 strong_best=1338 digest=3c61072b7f
lender_msa: 1.22s pairs=7333 strong_best=986 digest=336b6e4fb6
market: 4.86s pairs=7683 strong_best=2051 digest=ee656dbcf1
max_rss_mb 812.69921875
```

12 Same matching on a laptop (arm64)

Identical counts and digests to the s390x run.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % .venv/bin/python -I -c "import sys;sys.path.insert(0,'.');exec(open('scripts/remote_ma
tching.py').read())" 2>&1 | grep -v Warn

arch arm64 rows 37491
lender: 0.37s pairs=7621 strong_best=1338 digest=3c61072b7f
lender_msa: 0.70s pairs=7333 strong_best=986 digest=336b6e4fb6
market: 2.19s pairs=7683 strong_best=2051 digest=ee656dbcf1
```

13 XGBoost inside the LinuxONE container

XGBoost 3.4.1 in the container. Model sensitivity output saved there.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % ssh -i wise_key.pem linux1@148.100.109.157 'docker exec notebook /opt/venv/bin/python
-c "import xgboost;print(\"xgboost\",xgboost.__version__)"; docker exec notebook ls -la /home/ibm-user/secondlook/data/processed/model_s
ensitivity_xgb.json'

xgboost 3.4.1
-rw-r--r-- 1 ibm-user ibm-user 1829 Oct 17 14:33 /home/ibm-user/secondlook/data/processed/model_sensitivity_xgb.json
```

14 Validated Granite notes

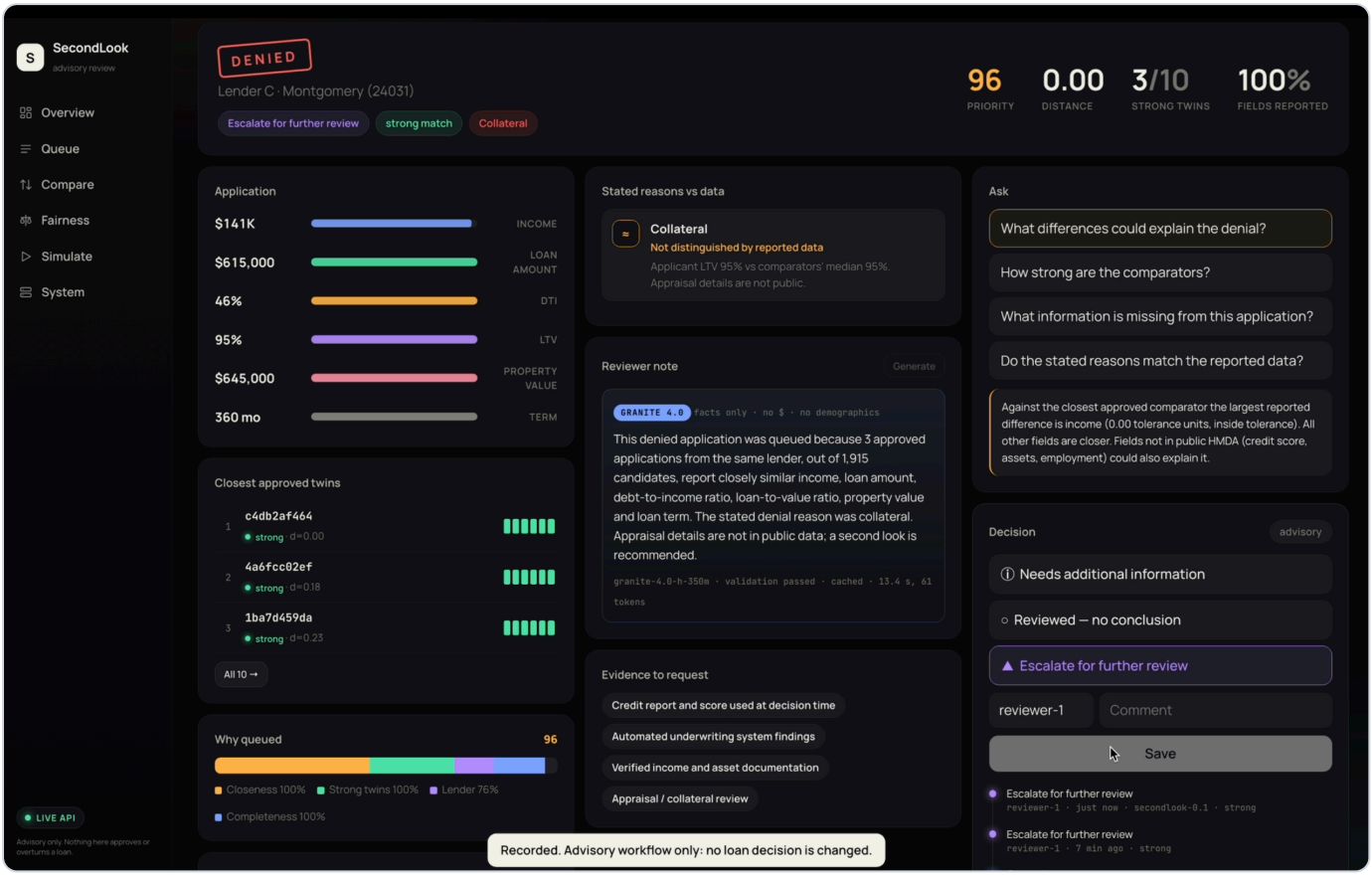
26 reviewer notes generated by Granite, validated and cached.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % ssh -i wise_key.pem linux1@148.100.109.157 'python3 -c "import json;c=json.load(open(\
"/home/linux1/secondlook/data/processed/notes_cache.json\"));print(len(c),\"validated Granite notes cached\")"'

26 validated Granite notes cached
```

15 Dashboard: case review

Reason check, validated Granite note, grounded questions, advisory decision with audit trail.



16 Automated tests

9 tests passed.

```
(.venv) shreyv@Shreys-MacBook-Pro WISE_Datathon % .venv/bin/python -m pytest -q tests

.........
9 passed in 2.31s
```

17 JupyterLab on the VM

Lab notebooks, SecondLook folder and the Granite model file in the IBM-provided container.

